

MODEL QUESTION PAPER

TED (15) 4111

Name.....

Revision 2015

Reg. No.....

Fourth Semester Diploma Examination in Tool and Die Engineering

MATERIAL TECHNOLOGY AND HEAT TREATMENT

(Maximum Marks: 100)

Time: 3 Hours

Part- A

(Maximum Marks: 10)

I. Answer the following questions in one or two sentences.

1. Define the term Unit Cell.
2. Select any two properties of Pig Iron.
3. List any two Aluminium Alloys.
4. State Visual Inspection.
5. Recognise Thermosetting Plastics.

(5 x 2= 10)

Part- B

(Maximum Marks: 30)

II. Answer any five questions.

1. Explain any three Mechanical Properties.
2. Derive the Atomic Packing Factor for BCC Structure.
3. Predict the effects on properties by any three alloying elements on Cast Iron.
4. Identify any three types of Brasses.
5. Differentiate Destructive and Non Destructive Testing
6. Compare Martempering and Austempering.
7. Classify Plastics with examples.

(5 x 6 = 30)

Part- C

(Maximum Marks: 60)

(Answer one full question from each unit)

UNIT –I

III. (a) Formulate Atomic Packing Factor for HCP Structure. (7)

(b) Illustrate the transition from liquid to solid state. (8)

OR

IV. (a) Compare Elastic and Plastic Deformation. (7)

(b) Differentiate Slip and Twinning Mechanism with neat sketch. (8)

UNIT- II

V. (a) Explain Open Hearth Process with neat sketch. (7)

(b) Construct Iron-Iron Carbide Phase Diagram (8)

OR

VI. (a) Explain Electric Arc Furnace with neat sketch. (7)

(b) Classify Alloy Steels. List properties and uses of any three alloy steels (8)

UNIT-III

VII. (a) Represent the Tensile Strength Test. (7)

(b) Explain Scanning Electron Microscope. (8)

OR

VIII. (a) Summarize on Aluminium Alloys. (7)

(b) Represent the Ultrasonic Test. (8)

UNIT- IV

IX. (a) Criticize about the defects in Heat Treatment Process. (7)

(b) Illustrate the Muffle Furnace. (8)

OR

X. (a) Interpret Polymerization and its Mechanisms. (7)

(b) Demonstrate TTT Diagram with sketch. (8)